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Deciphering the prodrome of inflammatory bowel disease up to 10 years before disease onset by massively parallel serology

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Conflicts of Interest

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Ethics Statement

The PREDICTS study was approved by the Institutional Review Board of the Naval Medical Research Center, Silver Spring, MD (NMRC.2014.0019) in compliance with all federal regulations governing the protection of human volunteers, and this research was performed under a Cooperative Research and Development Agreement (NMR 17–10209).

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Abstract

Background: Defining immune dysregulation during the asymptomatic prodrome of immune-mediated diseases offers opportunities for early disease detection and interception. In inflammatory bowel disease (IBD), prodromal immune changes remain poorly characterized.

Objective: To define pre-clinical immunological alterations by characterizing longitudinal serum antibody repertoires using high-throughput phage-display immunoprecipitation sequencing (PhIP-Seq).

Design: We applied PhIP-Seq to profile antibody responses in 2,000 longitudinal serum samples from 200 individuals who developed Crohn's disease (CD), 200 who developed ulcerative colitis (UC), and 100 matched healthy controls (HCs) within the US military PREDICTS cohort, collected up to 10 years before diagnosis. Antibody repertoires were profiled against 357,000 microbial-, viral-, food-, and immune-associated peptides.

Results: Antibody repertoire variability was increased up to ~4 years pre-diagnosis in pre-CD and pre-UC individuals. Differential analyses revealed elevated herpesvirus-directed responses (notably Epstein-Barr virus) and anti-flagellin antibodies up to 10 years pre-diagnosis in CD, particularly in individuals who later developed complicated or ileal disease. In contrast, responses to encapsulated bacteria (e.g., *Streptococcus pneumoniae*, *Haemophilus*, *Neisseria*) progressively declined towards diagnosis. Pre-UC was characterized by combined antimicrobial, antiviral, and autoantibody signatures, including antibodies against the MAP kinase-activating death domain (MADD) protein.

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Conclusion: Large-scale serological profiling of archived pre-diagnostic samples identified disease-specific immune trajectories years before IBD onset, providing novel insights into disease pathogenesis in its prodromal phase.

Keywords

Inflammatory bowel disease; antibodies; antibody repertoires; diagnosis; pre-diagnostics

Introduction

Chronic immune-mediated inflammatory diseases (IMIDs) such as inflammatory bowel diseases (IBDs), encompassing Crohn's disease (CD) and ulcerative colitis (UC), are believed to arise from a prolonged prodromal phase, during which immunological alterations accumulate years before clinical onset [1–4]. In this phase, disease-specific humoral immune responses have been reported, including anti-*Saccharomyces cerevisiae* antibodies (ASCA) and anti-flagellin antibodies (e.g., anti-Fla-X, anti-A4-Fla2) preceding CD, and perinuclear anti-neutrophil cytoplasmic antibodies (pANCA) or anti-αvβ6 antibodies preceding UC [3,5–10]. However, conventional assays capture only a narrow spectrum of antibody reactivities against pre-defined antigens, limiting insights into the breadth and dynamics of immune responses during disease development [11,12].

Phage-display immunoprecipitation sequencing (PhIP-Seq) enables massively parallel profiling of antibody repertoires against hundreds of thousands of peptides, spanning microbial, viral, and host-derived antigens [13–15]. This approach has been successfully applied to the microbiome [16] and to diverse disease contexts, including viral infections [17,18], autoimmune diseases [19–21], and IBD [22,23]. By interrogating immune responses at unprecedented scale, PhIP-Seq provides a powerful platform to delineate disease prodromes and immune trajectories. It has also provided insights into Epstein-Barr virus (EBV) immunity and its role in immune-mediated diseases [19,20,24].

Herein, we leverage PhIP-Seq in a large pre-diagnostic cohort – the PREDICTS (Proteomic Evaluation and Discovery in an IBD Cohort of Tri-service Subjects) cohort [25], comprising thousands of longitudinal serum samples collected up to a decade prior to diagnosis. Using IBD as a model, we identify antibody signatures that precede clinical onset and characterize their persistence and temporal trajectories. Our findings illustrate that large-scale serological profiling in pre-diagnostic IBD biobanks can illuminate prodromal immune alterations.

Materials and Methods

Study cohort and design

This nested case-control study was performed using data and biomaterials from the Proteomic Evaluation and Discovery in an Inflammatory Bowel Disease Cohort of Tri-service Subjects (PREDICTS) – a longitudinal pre-diagnostic IBD cohort that was established in collaboration with the United States Department of Defense Serum Repository (DoDSR) [25]. The PREDICTS study was approved by the Institutional Review Board of the Naval Medical Research Center, Silver Spring, MD (NMRC.2014.0019) in compliance

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with all federal regulations governing the protection of human volunteers, and this research was performed under a Cooperative Research and Development Agreement (NMR 17–10209). Using the Defense Medical Surveillance System (DMSS), new-onset cases of CD and UC were identified in the period of 1998–2013, which were defined as having two or more medical encounters with the International Classification of Diseases (ICD, 9th Revision, Clinical Modification – ICD-9-CM, https://www.cdc.gov/nchs/icd/icd9cm.htm, Accessed June 24, 2024) code for CD and UC (555.0, 555.1, and 555.9 for CD and 556 for UC, including all subgroup codes for both, which were linked to the DoDSR). Case identification followed the previously published PREDICTS/DMSS algorithm [25], requiring ≥2 IBD-coded medical encounters (to reduce misclassification based on single administrative entries) and ≥1 medical encounter with a CPT code for a lower gastrointestinal endoscopic procedure preceding or accompanying an IBD-coded encounter. The use of ≥2 ICD-9-CM codes as a case definition for IBD has previously been validated in military and other administrative databases, demonstrating positive predictive values of approximately 80–90% for Crohn's disease and ulcerative colitis [26–28]. Prior PREDICTS analyses additionally showed no significant differences in gastrointestinal-related ICD-9 coding between cases and controls in the two years preceding the index diagnosis date, supporting the validity of incident case identification [3]. In total, longitudinal serum samples from 200 individuals who developed CD, 200 individuals who developed UC, and 100 healthy individuals who did not develop IBD were selected from the DMSS and included in the present study. These individuals were selected from the larger PREDICTS cohort, comprising approximately 1,000 incident CD and 1,000 incident UC cases with linked longitudinal serum samples. From this cohort, individuals fulfilling the predefined inclusion criteria and with availability of longitudinal serum samples were first identified. From this eligible subset, 200 CD and 200 UC individuals were subsequently randomly selected for inclusion in the present study. The date of IBD diagnosis was defined as the first time an IBD ICD-9-CM code was registered. Healthy controls were selected from the same underlying source population within the DMSS and were frequency-matched to cases based on age, sex, race, and year of sample A collection (±1 year). This matching was performed at the level of group distributions rather than individual case-control pairs. More specifically, controls were required to have no ICD-9-CM codes for CD (555.0, 555.1, 555.9) or UC (556, including subgroup codes), and no codes for rheumatoid arthritis (714.0), celiac disease (579.0), or colorectal cancer (153.0–154.1), consistent with the original PREDICTS control selection criteria [25]. A smaller control group was included (ratio 2:1) because healthy controls primarily served as a reference population for background temporal variation, which enabled characterization of disease-specific longitudinal immune trajectories preceding IBD onset. For each study participant, four serum samples were obtained (Fig. 1A): around the time of diagnosis (± one year, with 10.5% collected prior and 89.5% after the index diagnosis date for individuals who developed CD, with a median [IQR] number of days relative to diagnosis of 243 [94–435] days; for UC and HC, samples were obtained precisely at the time of diagnosis, defined as sample A/diagnostic timepoint), followed by sequential samples obtained approximately 2 years prior to sample A (sample B, ~2Y PD) and approximately 4 years prior to sample A (sample C, ~4Y PD), consistent with the structured surveillance sampling of the DoDSR. Sample D represented the earliest available archived serum sample for each individual,

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typically corresponding to approximately 10 years prior to diagnosis (~10Y PD). Serum samples previously archived at −80°C were randomized on plates by sample timepoint and clinical diagnosis prior to any measurements, thereby minimizing potential bias arising from differential storage duration. Basic demographic data, including age, sex, and race, were obtained from all participants. Information on time of symptom onset or medication use was not available. Disease phenotypes, including disease location, disease behavior, and surgical history were collected from individuals who developed IBD and ascertained based on ICD-9-CM codes. Disease location and behavior were classified according to the Montreal classification.

Patient and public involvement

Patients and/or the public were not involved in the design, or conduct, or reporting, or dissemination plans of this research.

PhIP-Seq procedures

Design and content of the antigen libraries: Details on the design, content, antigen processing, and cloning of the PhIP-Seq libraries can be found in the Supplementary Methods. Three antigen libraries covering a total of 357,000 different antigens (Fig. 1B) were used in this study: a 244,000-variant library (“agilent”) [16], a 100,000-variant library (“twist”) [29], and a 13,000-variant coronavirus library (“corona2”) [30].

The 244,000-variant library encompasses a wide array of microbial antigens derived from 28,668 different proteins, including 27,837 proteins from microbiota antigens. It features antigens from various sources: bacterial genes (122,551 peptides, ~50%) and strains (24,510 peptides, ~10%) of commensal bacteria selected from metagenomic sequencing data of nearly 1,000 healthy individuals [31], selected pathogenic bacteria (24,500 peptides, ~10%) [16], probiotic bacteria (14,700 peptides, ~6%) [32], known antibody-coated bacteria (22,050 peptides, ~9%) [33], virulence factor antigens (24,164 peptides, ~10%) from the virulence factor database (VFDB) [34], and a set of biological and technical controls (11,525 peptides, ~5%, see [16] for details). In addition to these standard subgroups, a custom subgroup of flagellin peptides was also created based on a previously published annotation [35].

The 100,000-variant library comprises antigens from bacteriophages (~40,000 peptides, selected from those infecting commensal bacteria, probiotic bacteria, and pathogenic bacteria) [29], allergens from six different allergen databases (~30,000 peptides) [36–41], and all B cell antigens as deposited in the Immune Epitope Database (IEDB) on January 14th, 2018, except for non-human host, allergens (which were taken separately), and Chagas disease epitopes (~30,000 peptides) [42]. Allergen peptides were derived from the IEDB list [41], the WHO/IUIS Allergen Nomenclature Database (www.allergen.org) [36], AllergenOnline (www.allergenonline.org) [37], Allergome (www.allergome.org) [38], the Database of allergen families, AllFam (www.meduniwien.ac.at/allfam/) [39], the Structural Database of Allergenic Proteins (SDAP) (femi.utmb.edu) [40], and from allergenic helminthic proteins [41]. To avoid redundancy, antigen sequences that appeared in more than one database were included only once.

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The 13,000-variant coronavirus-focused library was created following the COVID-19 pandemic and contains antigens from both seasonal/common cold human and animal coronaviruses. Further details on the creation of this antigen library can be found in [30].

For all antigen libraries, antigens were rationally selected by prioritizing previously reported bacterial antigens and strains that are known to be recognized by the immune system as well as uncharacterized antigens. Here, the focus was primarily put on their functions, particularly those with a high likelihood of being exposed to the immune system, such as membrane-, surface-, motility-, and secreted proteins (see [16] for details). Antigen selection was conducted without considering peptide sequence length, necessitating peptide splitting within the given 230 nt limit of DNA oligonucleotide synthesis. Further details on the antigen library composition and antigen selection process can be found elsewhere [16].

Immunoprecipitation and next-generation sequencing: Experimental details of the immunoprecipitation (IP) and next-generation sequencing (NGS) steps can be found in the Supplementary Methods.

Antibody testing using a commercially available panel (Prometheus Panel): Each serum sample analyzed for this study was also previously tested [3] for a specified panel of circulating serological antibodies, including anti-*Saccharomyces cerevisiae* antibodies (ASCA, both IgG and IgA), perinuclear anti-neutrophil cytoplasmic antibodies (pANCA), anti-*Escherichia coli* outer membrane porin C (anti-OmpC), and three distinct anti-flagellin antibodies (anti-CBir1, anti-Fla-X, and anti-A4-Fla2). These antibodies were measured using commercially available ELISAs performed at Prometheus Laboratories (San Diego, CA, USA).

Statistical analysis

Processing of raw PhIP-Seq data: Raw sequencing reads obtained from NGS after IP were reduced to 1.25 million identifiable reads per sample, only capturing those within one error of all possible barcodes for which the paired-end matched the identifiable antibody-bound peptide. For data analysis, at least 750,000 reads were required in case fewer reads were obtained. Antigen enrichment was calculated through comparing the total number of reads per antigen with the input read numbers, which were determined when the phage library was sequenced before IP. Each input read number per sample was assumed to generate an output read number null distribution that was fitted using a generalized Poisson distribution. Parameters were estimated separately for each input read number in each sample, and scores were subsequently generated by fitting the parameters to three distribution parameters for all samples, followed by interpolation for each input read number [14]. Fold changes were calculated as the number of reads for antibody-bound peptides (after IP) vs. the number of input reads (prior to IP), computed only for significantly enriched antibody-bound peptides (which were referred to as seropositive antibodies), while the remaining peptides were set to zero. Additionally, fold changes were calculated only with a minimum of 25 input reads per peptide. Baseline sequencing of the antigen phage library (prior to IP) was conducted at >100-fold coverage. Samples with fewer than 200 significantly enriched antigens compared to input reads were excluded. For most statistical

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analyses (unless otherwise specified), prevalence thresholds were applied to filter for antibody-bound peptides that appeared in at least 5% but in equal or less than 95% of samples.

General statistical analyses: Baseline descriptive statistics of the study cohort were presented as means ± standard deviation (SD), medians with interquartile range [IQR] or as proportions with corresponding percentages (%). Normality was assessed through visual inspection of normal probability (Q-Q) plots and histograms. Differences across groups were tested using one-way analysis of variance (ANOVA) with post-hoc tests or nonparametrically using Kruskal-Wallis tests with post-hoc Mann-Whitney *U*-tests, as appropriate. Principal component analysis (PCA) was performed for dimensionality reduction purposes and to identify potential determinants of data clustering. Overall antibody diversity across samples was determined through calculating the Shannon diversity index/alpha-diversity, based on the number of antibody-bound peptides significantly bound. For longitudinal analysis of antibody-bound peptide counts across repeated sampling timepoints, global differences were first assessed using Friedman tests, followed by pairwise Wilcoxon signed-rank tests with Holm correction for multiple testing. Longitudinal stability of antibody repertoires across the different study timepoints was analyzed through calculating point-biserial or phi (φ)-coefficients or mean square contingency coefficients, which are suitable for analyzing correlations between binary variables. To analyze longitudinal changes in frequencies of antibody repertoires over time, generalized linear mixed-effects models with logit-link function were utilized to account for repeated measurements by applying random effects over individual subjects and to enable testing for interactions between study groups and timepoints. For case-control analyses, which were performed to identify differentially abundant antibody-bound peptides across groups at each study timepoint, logistic regression analysis with covariate adjustment was performed with consistent adjustment for the effects of age and sex. For all longitudinal and cross-sectional analyses, we analyzed data both on the level of individual antibody-bound peptides and on the level of antigen subgroups, for which a taxonomical classification dictionary was used based on at least 5 peptides that occurred within 5–95% of samples within a given analysis (see **Table S86**). For selected structural analysis of antibody-bound peptides, we performed multiple sequence alignments aiming to identify potentially shared motifs, using MEGA software with the multiple sequence comparison by log-expectation (MUSCLE) algorithm in default settings [43]. Subsequent visualization of these alignments including the consensus sequence, conservation %, and sequence logos were performed using CLC Main Workbench 6 software. To study associations between Prometheus panel antibodies [3] and PhIP-Seq-derived antibody-bound peptides, hierarchical clustering analysis with the complete linkage method was used, where a pairwise distance matrix was computed using the Euclidean distance metric. Clusters of Prometheus antibodies and PhIP-Seq-derived antibodies were defined by applying a threshold distance set at 20% of the maximum pairwise distance. Multiple testing adjustment was performed adopting a false discovery rate (FDR) <0.05 according the Benjamini-Hochberg method. Data analysis and data visualization were performed using Python (v.3.8.5, Python Software Foundation), using the following libraries: *pandas* (v.1.5.2), *numpy* (v.1.24.0), *statsmodels* (v.0.13.2),

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scipy (v.1.8.1), *scikit-learn* (v.1.0.2), *matplotlib* (v.3.4.3), and *seaborn* (v.0.11.1). Non-data visualizations were made using BioRender.

Results

Study cohort, overall characterization and longitudinal stability of antibody repertoires up to 10 years before IBD diagnosis

We analyzed 2,000 longitudinal serum samples at ~10-, ~4-, and ~2 years before diagnosis, and shortly after IBD diagnosis (Fig. 1A). Antibody repertoires were profiled against 357,000 rationally selected peptide antigens from gut microbiota, viruses, phages, allergens, and self-antigens (Fig. 1B) [16,22]. Study groups were matched for age (at diagnosis) (CD: 33.8±6.1; UC: 33.5±5.5; HC: 33.5±5.5 years, P=0.699, one-way ANOVA), sex (%females: CD 12.5%, UC 8.0%, HC 8.0%, P=0.253, chi-square) and race (P=0.103, Fisher's exact) (Fig. 1C, Table 1), minimizing known confounders affecting antibody repertoires [16].

In total, 123,980 unique antibody-bound peptides were detected in at least one sample, with most responses present in only few individuals, consistent with previous observations [16,22]. Analyses focused on 2,156 peptides present in ≥5% of samples (Suppl. Fig. S1A), unless stated otherwise. At the individual level, antibody diversity (Shannon) was significantly lower in pre-UC (median 727 [IQR: 416–1,054] vs. HCs 982 [IQR: 665–1,195] peptides) at diagnosis and ~2Y PD, and lower in pre-UC vs. pre-CD (median 835 [IQR: 557–1,156]) at diagnosis (Fig. 1D). Throughout the pre-diagnostic phase, antibody distributions per individual were comparable across groups (Suppl. Fig. S1B–E), However, bound peptide counts significantly decreased at diagnosis in CD (P=0.002, Friedman test) and UC (P<0.001, Friedman test) compared to ~4Y PD (CD: P=0.02; UC: P=0.01, Wilcoxon signed-rank tests, Suppl. Fig. S1F,G), but did not change in HCs (P=0.97, Friedman test) (Suppl. Fig. S1H, **Table S1**).

To assess the impact of diagnosis and timepoints on antibody repertoires, we performed principal component analysis (PCA) for dimensionality reduction (Suppl. Fig. S1I–R). Comparing the first ten principal components (PCs) across individuals developing CD, UC, and HCs (Suppl. Fig. S1I, L–O), CD and UC cases differed significantly from HCs in seven of ten PCs (all P<0.001, Kruskal-Wallis followed by post-hoc tests), with a cumulative explained variance of 20%. Labeling by study timepoint (Suppl. Fig. S1J, P–R) revealed a subtle trajectory from the earliest pre-diagnostic timepoint (~10Y PD) towards diagnosis, with seven of ten PCs significantly shifting over time (all P<0.05, Wilcoxon tests). PCA of the first two PCs (Suppl. Fig. S1K) identified separation driven by peptides from cytomegalovirus (CMV) (PC2) and *Streptococcus* (PC1) (**Table S2**). Similar patterns were observed in PCA performed separately by each study timepoint (Suppl. Fig. S1L–O) and group (Suppl. Fig. S1P–R), suggesting that these PC shifts likely reflect time-dependent rather than disease-induced changes.

We correlated paired antibody repertoires at diagnosis with pre-diagnostic samples (~10Y, ~4Y, and ~2Y PD) in individuals developing CD, UC, and HCs (Suppl. Fig. S2A–C, Suppl. Fig. S3). Comparing phi coefficient distributions (Suppl. Fig. S2D–F) showed that at ~4Y PD, individuals who developed CD had lower similarity to their diagnostic antibody

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repertoires than those who developed UC or HCs (Fig. 1E, adjusted $P<0.001$, Wilcoxon), which was not observed at ~10Y PD (Suppl. Fig. 2G). At ~2Y PD, individuals developing UC showed lower similarity than HCs, followed by those developing CD (Suppl. Fig. 2H, adjusted $P<0.001$ and $P<0.05$, respectively). These observations indicate differences in antibody repertoire similarity in pre-CD and pre-UC individuals compared to HCs up to ~4Y PD.

Longitudinal analysis from the earliest pre-diagnostic stage to the peri-diagnostic timepoint reveals increased antibody reactivity against herpesviruses and decreased reactivity against potentially pathogenic bacteria and respiratory viruses

To explore disease-specific changes, we analyzed antibody responses at epitope-level resolution. Longitudinal changes over ~10 years were assessed *within* each group (pre-CD, pre-UC, HCs; Fig. 2, Suppl. Fig. S4). Because the study design includes a (peri-)diagnostic reference sample (sample A), longitudinal analyses were used to contextualize how antibody repertoires evolve from the earliest pre-diagnostic timepoint (~10Y PD) to diagnosis. Importantly, the peri-diagnostic sample serves as an internal reference that enables assessment of within-individual trajectory and repertoire stability over time. These analyses therefore provide context for the pre-diagnostic findings rather than implying that changes observed close to diagnosis are predictive themselves. While few peptides differed significantly after multiple testing correction, differences were observed at the antigen subgroup level (**Tables S3-S20**). Statistical analysis confirmed distinct shifts in taxonomic subgroups per group. In individuals who developed CD, antibody responses against EBV (antibody prevalence ratio [APR] 1.27, adjusted $P=3.0\times10^{-26}$), CMV (APR 1.54, adj. $P=2.4\times10^{-24}$), HSV-1 (APR 1.31, adj. $P=1.6\times10^{-7}$), HSV-2 (APR 2.14, adj. $P=1.6\times10^{-4}$), *Pseudomonas aeruginosa* (APR 1.38, adj. $P=1.5\times10^{-3}$), and RSV (APR 1.17, adj. $P=1.6\times10^{-2}$) were elevated at diagnosis vs. ~10Y PD (Fig. 2A,D, **Table S3**). Conversely, responses against *Streptococcus* (APR 0.88, adj. $P=1.7\times10^{-8}$), *Haemophilus influenzae* (APR 0.81, adj. $P=1.9\times10^{-4}$), *Neisseria meningitidis* (APR 0.72, adj. $P=7.9\times10^{-3}$), and enteroviruses (type C) (0.85, adj. $P=0.04$) declined.

In pre-UC, higher antibody frequencies at diagnosis vs. ~10Y PD were observed for EBV (APR 1.33, adj. $P=3.9\times10^{-34}$), CMV (APR 1.81, adj. $P=2.3\times10^{-25}$), HSV-1 (APR 1.32, adj. $P=3.4\times10^{-8}$), influenza (APR 1.15, adj. $P=6.0\times10^{-8}$), HSV-2 (APR 1.97, adj. $P=7.5\times10^{-4}$), *Enterobacteriaceae* (APR 1.19, adj. $P=0.01$), *Escherichia coli* (APR 1.04, adj. $P=0.01$), *Haemophilus parainfluenzae* (APR 1.02, adj. $P=0.03$), and *P. aeruginosa* (APR 1.16, adj. $P=0.04$) (Fig. 2B,E, **Table S4**). Reduced frequencies of antibody responses were observed for rhinoviruses (APR 0.67, adj. $P=2.1\times10^{-5}$), *H. influenzae* (APR 0.76, adj. $P=2.3\times10^{-3}$), coxsackieviruses (APR 0.66, adj. $P=3.5\times10^{-3}$), enteroviruses type A71 (APR 0.62, adj. $P=0.01$) and C (APR 0.81, adj. $P=0.02$), Bacteroidales (APR 0.82, adj. $P=0.05$), *Legionella pneumophila* (APR 0.73, adj. $P=0.05$), *Lachnospiraceae* (APR 0.72, adj. $P=0.05$), *Blautia* (APR 0.79, adj. $P=0.05$), and adenoviruses (APR 0.75, adj. $P=0.05$) when compared to ~10Y PD.

In matched HCs, increases were observed for EBV (APR 1.28, adj. $P=6.3\times10^{-13}$), CMV (APR 1.27, adj. $P=4.1\times10^{-9}$), influenza (APR 1.40, adj. $P=7.1\times10^{-5}$), HSV-1 (APR 1.38,

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adj. P=7.5×10^{-5}), HSV-2 (APR 1.88, adj. P=4.7×10^{-3}), and *Staphylococcus* (APR 1.07, adj. P=0.02) at the diagnostic timepoint vs. ~10Y PD (Fig. 2C,F, **Table S5**). Decreases were observed for *Streptococcus* (APR 0.93, adj. P=5.9×10^{-8}), *H. influenzae* (APR 0.88, adj. P=9.9×10^{-4}), and rhinoviruses (APR 0.92, adj. P=0.04). Although significant, changes were smaller than in CD or UC, particularly for EBV. Most trends persisted across ~4Y PD and ~2Y PD (Suppl. Fig. S4), with few exceptions limited to ~10Y PD vs. diagnosis (**Tables S6-S11**).

To explicitly characterize longitudinal antibody trajectories across the pre-diagnostic phase, we analyzed antibody responses across four timepoints (~10Y, ~4Y, ~2Y PD, and diagnosis) using generalized logistic mixed-effects models. Here, we identified 66 and 55 unique antibodies with significant longitudinal changes in pre-CD and pre-UC, respectively (**Tables S21-S22**). In pre-CD, escalating trajectories were observed for antibodies against flagellins (e.g., *Roseburia*), EBV (VP26), and CMV (glycoprotein B), while responses to *Neisseria* (IgA1 protease) declined (Fig. 3A).

Among pre-UC, escalating trajectories were observed for antibodies targeting MADD, and decreases for antibodies against enteroviruses (type C), *Staphylococcus* (fibronectin-binding proteins, aureolysins), and *H. influenzae* (HxuA) (Fig. 3B).

Overall, these findings reveal increasing antibody reactivity against herpesviruses and declining reactivity against encapsulated bacteria and respiratory viruses in individuals developing IBD.

Pre-clinical IBD is marked by distinct antibody responses already up to 10 years before diagnosis

Next, we studied group-specific differences in antibody-bound peptide frequencies, comparing individuals who developed CD (Fig. 3C–D) and UC (Fig. 3E–F) with matched HCs, as well as CD vs. UC (Fig. 3G–H). Analyses were performed per timepoint across antigen subgroups, individual bound peptides, and default library subgroups (Fig. 3, Suppl. Figs. S5–S7 and **Tables S23-S36**).

In pre-CD, antibodies against EBV (APR 1.14, adj. P=6.6×10^{-9}, EBNA-1–6, BMRF1/BZLF1), CMV (APR 1.11, adj. P=1.2×10^{-3}, phosphoprotein 150), and influenza (APR 1.20, adj. P=2.0×10^{-3}, hemagglutinin peptides) were overrepresented vs. HCs (Fig. 3C–D, **Table S23**). Antibodies targeting *Streptococcus* (APR 0.89, adj. P=1.5×10^{-3}), rhinoviruses (APR 0.82, adj. P=1.9×10^{-3}), Bacteroidales (APR 0.93, adj. P=9.8×10^{-3}, e.g., *Parabacteroides merdae*), *Helicobacter pylori* (APR 0.67, adj. P=0.01), *Blautia* (APR 0.75, adj. P=0.02) and most *Staphylococcus aureus* antigens (fibronectin-binding proteins, Ig-binding proteins, IsdB) were underrepresented (Fig 3C–D, Suppl. Fig. S5a–f, Suppl. Fig. S6a–d, **Table S23**).

At the peptide level, overrepresented responses targeted *S. aureus* (lipase 2, autolysins), *P. merdae* (uncharacterized proteins), and *H. influenzae* (adhesins); underrepresented responses included Clostridiales (peptidase S11), *Lactobacillus* (N-acetylmuramoyl-L-alanine amidase), and enteroviruses A71, B, C (**Tables S24-S27**). Default library subgroup analysis (Fig. 1B, Suppl. Fig. S7, **Table S28**) revealed underrepresentation of VFDB, IEDB,

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pathogenic bacteria, and gut microbiome antigens, but overrepresentation of flagellins (mainly *Lachnospiraceae*), allergens, and IEDB antigens. Dominant anti-flagellin antibody responses targeted the N-terminal 'hinge' epitope (nD0–$1_{p25–59}$), previously implicated as a prominent B cell target in CD and healthy infants [44], largely derived from *Lachnospiraceae*, (Suppl. Figs. S8–S9), and strongly exceeding responses to pathogenic species (*Salmonella*, *Shigella*, *Yersinia* (Suppl. Fig. S8b, **Table S37**).

In pre-UC, antibodies against *Lactobacillus* (APR 1.09, adj. P=0.02 at ~10Y PD, APR 1.08, adj. P=1.3×10^{-3} at ~2Y PD, APR 0.97, adj. P=1.2×10^{-3} at diagnosis), CMV (APR 1.05, adj. P=2.0×10^{-3} at ~4Y PD, APR 1.08, adj. P=4.7×10^{-3} at ~2Y PD, APR 1.09, adj. P=2.2×10^{-10} at diagnosis), HSV-1 (APR 1.13, adj. P=0.05 at ~4Y PD), and *Enterobacteriaceae* (APR 1.09, adj. P=0.01 at diagnosis) were overrepresented, while antibodies against rhinoviruses (APR 0.74, adj. P=0.01 at diagnosis) were underrepresented (Fig. 3E–F, Suppl. Fig. S5g–l, Suppl. Fig. S6e–h, **Table S23**). Peptide-level analyses (**Tables S29-S32**) showed overrepresentation of antibodies against *S. aureus* (lipase 2) and *Bacteroides* (TPR/NAGPA), and underrepresentation of responses to *S. aureus* (fibronectin-binding proteins, aureolysin), *S. pneumoniae* (choline-binding proteins), *S. pyogenes* (C5a peptidase), Clostridiales (FAD:protein FMN transferase), *Enterobacteriaceae* (outer membrane proteins), *H. influenzae* (adhesins), and *Bacteroides* (membrane proteins). Responses to VZV, HSV-1, influenza, and enteroviruses (coxsackievirus A9) were also underrepresented. At diagnosis, MADD-targeted antibody responses were strongly overrepresented. Default library subgroup analysis revealed overrepresentation of flagellins (~4Y PD, ~2Y PD, diagnosis), probiotic bacteria (all timepoints), and allergens (~10Y PD), and underrepresentation of VFDB (~4Y PD, ~2Y PD, diagnosis), coronaviruses (~2Y PD), antibody-coated bacteria (~4Y PD), and IEDB antigens (~10Y PD and ~4Y PD) (Suppl. Fig. S7, **Table S28**).

In CD vs. UC comparisons, CD-associated responses to EBV (APR 1.36, adj. P=1.3×10^{-16}) and CMV (APR 1.45, adj. P=2.1×10^{-12}) were stronger at ~10Y PD and persisted across timepoints (~4Y PD: APR 1.19, EBV adj. P=7.7×10^{-3}; ~2Y PD: APR 1.33, EBV adj. P=3.6×10^{-18}, CMV adj. P=2.7×10^{-6}; diagnosis: EBV APR 1.30, adj. P=2.7×10^{-9}, CMV APR 1.19, adj. P=2.1×10^{-3}) (Fig. 3G–H, Suppl. Fig. S5M–R, Suppl. Fig. S6I–L, **Table S23**). Pre-CD was also associated with higher responses to *H. influenzae* (~4Y PD, APR 1.25, adj. P=1.0×10^{-4}), *Staphylococcus* (diagnosis, APR 1.02, adj. P=1.4×10^{-5}), and *Streptococcus* (diagnosis, APR 1.04, adj. P=1.9×10^{-3}), while UC was associated with elevated responses to rhinoviruses (~10Y PD, APR 0.90, adj. P=5.8×10^{-3}), HSV-1 (APR 0.96, ~4Y PD, adj. P=0.02), Bacteroidales (APR 0.98, ~2Y PD, adj. P=6.1×10^{-3}), *Lactobacillus* (~2Y PD, APR 0.93, adj. P=0.01; diagnosis, APR 1.00, adj. P=0.03) (Fig 3G–H, **Tables S23, S33–36**). Default library subgroup comparisons revealed prominent overrepresentation of responses against flagellins (**Table S28**), allergens (all timepoints), IEDB (~10Y PD, ~2Y PD, diagnosis), gut microbiome (~10Y and ~2Y PD), VFDB (~10Y PD, ~2Y PD, diagnosis), and pathogenic bacterial (~4Y PD, ~2Y PD, diagnosis) antigens (Suppl. Fig. S7).

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Altogether, these data reveal antibody repertoire differences among individuals developing IBD, detectable up to ~10Y PD, with prominent EBV- and flagellin-directed responses in pre-CD vs. pre-UC and HCs.

Antibody responses against EBV target distinct sequence motifs of BMRF1, EBNA-1, and EBNA-6 proteins

Given the predominant overrepresentation of anti-EBV antibodies in pre-diagnostic CD (vs. HCs and future UC), we further investigated their antigenic specificity and identified potential shared epitopes (Fig. 4). We assembled all overrepresented EBV-directed peptides from case-control analyses (Fig. 3, **Tables S24-S27**, **S29-S36**, **Table S38**), which grouped into three antigen categories: lytic proteins (BMRF1, BZLF1), latent proteins (EBNA-1 to −6), and capsid proteins (VP26).

Antigen-specific comparisons of anti-EBV antibody frequencies revealed significant overrepresentation in pre-diagnostic CD, particularly for BMRF1 (Fig. 4B, ~2Y PD and diagnosis vs. HCs), as well as BZLF1 (~2Y PD), EBNA-1 (~10Y PD), EBNA-2 (~2Y PD and diagnosis), EBNA-4 (diagnosis), and VP26 (~10Y PD) when compared to future UC (Fig. 4D). Such differences were absent when comparing future UC vs. HCs (Fig. 4C),indicating that the strongest EBV-associated signals were enriched in pre-diagnostic CD, although several of these differences were more pronounced in future CD vs. future UC comparisons than in direct case-control comparisons.

Because proteins are tiled into overlapping 64- and 54-aa peptides for PhIP-Seq library construction, we examined sequence overlap among EBV-targeted peptides. Multiple sequence alignment (**Table S38**, Fig. 4E, Suppl. Fig. S10) revealed three distinct clusters derived from EBNA-1, BMRF1, and EBNA-6, each sharing conserved motifs. For example, although EBNA-1-antibodies did not differ systematically between groups, many targeted aa 408–448, within the 381–452) region recently identified as a dominant EBNA-1 epitope in MS [45], which includes the 411–426 domain cross-reactive with MBP in MS [46].

Overall, pre-diagnostic anti-EBV antibody responses converge on specific motifs within BMRF1, EBNA-1, and EBNA-6, possibly reflecting shared immunogenicity or molecular mimicry, as described for the EBNA-1 381–452 region in MS [20].

Complicated CD and CD disease location exhibit distinct antibody signatures and can be accurately predicted up to 10 years before diagnosis

Identifying patients at risk for complicated CD is clinically important, as these individuals typically require distinct treatment strategies [1]. Some studies suggest that certain serological markers may predict complicated CD at diagnosis [7]. Here we investigated whether such antibody signatures can also be identified through our analyses (**Tables S39-S43**).

To this end, we compared antibody repertoires across all timepoints between individuals who developed complicated (n=56, defined by obstructing and/or penetrating disease and/or surgical history) and non-complicated CD (n=144), analyzing both antigen subgroup levels

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(**Table S39**) and individual bound peptides while adjusting for age and sex (**Tables S40-S43**).

Among those who developed complicated CD, samples obtained ~10Y PD showed a significant overrepresentation of antibody responses against EBV (Fig. 4i) (adj. $P=2.9\times10^{-11}$), *Eubacterium* (adj. $P=0.03$, primarily flagellins), and influenza (adj. $P=0.04$) compared with non-complicated CD. At subsequent pre-diagnostic timepoints, responses to EBV (~4Y PD, adj. $P=1.2\times10^{-9}$; ~2Y PD, adj. $P=1.7\times10^{-7}$), *Eubacterium* (~2Y PD: adj. $P=4.6\times10^{-3}$, mainly flagellins), as well as CMV (~2Y PD, adj. $P=3.9\times10^{-3}$) and *Roseburia* (~2Y PD, adj. $P=0.02$, driven by flagellins) were overrepresented. Conversely, responses to *H. pylori* (~4Y PD: adj. $P=2.5\times10^{-5}$; ~2Y PD, adj. $P=0.04$), *Staphylococcus* (~2Y PD, adj. $P=4.6\times10^{-3}$), *Salmonella* (~2Y PD, adj. $P=0.01$), influenza (~2Y PD, adj. $P=0.02$), *Campylobacter* (~2Y PD, adj. $P=0.02$), *H. parainfluenzae* (~2Y PD, adj. $P=0.02$), and *Shigella* (~2Y PD, adj. $P=0.02$) were underrepresented.

Similarly, we investigated whether antibody repertoires could discriminate CD disease location at diagnosis, focusing on colonic CD (Montreal L2), a subtype often considered a distinct clinical entity [S11]. Comparing individuals who developed colonic CD (L2, $n=44$) to those with ileal or ileocolonic CD (L1/L3, $n=102$), we observed significantly higher pre-diagnostic antibody responses to *Blautia* (~10Y PD, adj. $P=2.9\times10^{-4}$; ~4Y PD, adj. $P=0.02$) and influenza (~10Y PD, adj. $P=0.01$), and lower responses to EBV (~10Y PD, adj. $P=1.5\times10^{-5}$; ~4Y PD, adj. $P=0.03$; ~2Y PD, adj. $P=0.02$) and CMV (~10Y PD, adj. $P=1.2\times10^{-6}$; ~4Y PD, adj. $P=1.9\times10^{-4}$) in individuals who later developed colonic CD (**Tables S44–48**). A detailed overview of differentially abundant peptides between (pre-)colonic CD and (pre-)ileal/ileocolonic is provided in **Tables S45-S48**. Additional comparisons, including disease extent in UC (pancolitis vs. limited colitis) and perianal CD, are provided in **Tables S49-S56**.

Known serological biomarkers of IBD exhibit shared and divergent co-occurrences with PhIP-Seq-profiled antibody responses

Although the classical serological biomarkers of IBD from the Prometheus panel have previously shown predictive value for CD diagnosis [5–7,9], it is yet not fully described what immune reactivity associates with their presence and whether they may confer different specificities. As such, using available data from the same samples [3], we aimed to investigate what types of antibody responses co-occur with these serological markers and whether there are shared and/or specific signatures for each (Fig. 5). Unsupervised hierarchical clustering was performed on these Prometheus panel antibodies and PhIP-Seq-derived antibody-bound peptides that showed at least one association with an antibody across study timepoints (filtered on $P<0.01$ following logistic regression analysis, see **Tables S57-S84**) (Fig. 5A). While a dominant cluster of PhIP-Seq-derived bound peptides, almost exclusively consisting of anti-flagellin antibodies (representing clusters 6–11, full list in **Table S85**), co-occurred with ASCA (IgA and IgG) and anti-flagellin antibodies already at ~10Y PD (e.g., with anti-CBir1 levels, Fig. 5C), all antibodies demonstrated divergent co-occurrences with unique sets of PhIP-Seq-derived antibody responses (Fig. 5B–E). For example, at ~10Y PD, cluster 5 was predominantly composed of anti-EBV (particularly

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EBNA-1, Fig. 5D) and anti-CMV antibodies (e.g., phosphoprotein 150, Fig. 5E) that associated mostly with anti-OmpC levels (Fig. 5B,D,E). Cluster 4 showed a predominance of pANCA antibodies co-occurring with reduced antibodies against several enteroviruses (adenovirus C, enterovirus types B and C, coxsackievirus A9) and lower reactivity against *S. aureus* proteins (fibronectin-binding protein A, autolysins), *S. pneumoniae* (surface proteins, choline-binding proteins) and several Clostridiales/*Lachnospiraceae* proteins. Taken together, this analysis demonstrates both overlapping and divergent antibody reactivities among these antimicrobial antibodies that are well-known as serological markers for diagnosis. This data suggests they arise from different biological pathways underlying disease initiation and, thus, likely have different specificities.

Discussion

This study delineates a previously underappreciated landscape of serological responses to microbial and non-microbial antigens up to a decade before IBD diagnosis. Longitudinal analysis of 200 pre-CD, 200 pre-UC, and 100 HC individuals over approximately 10 years revealed relatively stable antibody repertoires throughout most of the pre-diagnostic phase, with increased variability closer to diagnosis. Further, we observed striking differences in antibody responses among individuals who developed CD or UC compared with matched HCs, characterized by elevated reactivity to herpesviruses (particularly EBV, CMV, and HSV-1/2) and specific bacterial antigens (e.g., *S. aureus*) closer to diagnosis. In contrast, reactivity to certain potentially pathogenic, often encapsulated bacteria (e.g., *Streptococcus*, *Haemophilus, Neisseria*) and viruses (e.g., rhinoviruses) decreased during the pre-diagnostic phase. Importantly, these data illustrate the power of massively parallel serological profiling to uncover dynamic immune trajectories years before disease onset, highlighting the potential of this approach for studying prodromal immune alterations in IBD and potentially other immune-mediated conditions. Notably, most of our key findings were already detectable at the earliest pre-diagnostic timepoint (~10 years before diagnosis), which strongly argues against peri-diagnostic or post-diagnostic effects driving the observed signals. This study provides proof-of-concept that prodromal antibody repertoires can serve as informative biomarkers in IBD, illustrating how early immunological shifts can be systematically captured and modeled in the context of IBD.

Elevated EBV-directed antibodies formed one of the strongest pre-diagnostic signatures for individuals who later developed CD, particularly those who developed ileal and complicated disease, detectable up to ~10 years before diagnosis. Three distinct and conserved epitopes within the BMRF1, EBNA-1, and EBNA-6 antigens were targeted. The EBNA-1 sequence motif, specifically within residues 381–452, corresponds to the site targeted by antibodies in patients with MS [45] which cross-reacts with MBP [46]. These observations should also be interpreted in the context of a previous PhIP-Seq analysis in established IBD [13], where EBV-associated differences were more strongly represented by lytic/reactivation-associated antigens such as BZLF1 and BMRF1, while EBNA-1 responses were among the few EBV-directed responses reduced in established CD vs. non-IBD controls as well as in recently diagnosed IBD vs. pre-diagnostic IBD [23]. The present pre-diagnostic findings extend these findings by suggesting that epitope-level convergence on EBNA-1 may already be detectable before diagnosis, even though EBNA-1 was not among the strongest direct case-control

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signals at the antigen level. In contrast, to our knowledge, the shared epitope motifs in BMRF1 and EBNA-6 have not previously been linked to inflammatory disease. A recent PhIP-Seq-based study identified a predictive MS signature comprising a cluster of antibodies with motifs resembling a variety of human pathogens, including EBV [19]. The role of EBV reactivity in MS is well-documented, supported by extensive epidemiological data [20], and is gaining recognition for its potential involvement in IBD pathogenesis, possibly through mechanisms such as molecular mimicry or loss of B cell tolerance. Severe EBV infections have been associated with an increased risk of developing IBD later in life [47], and EBV exposure was enriched in both pre- and post-diagnostic cohorts of patients with CD [24]. While the mechanisms linking anti-EBV antibodies to CD onset remain elusive, several hypotheses warrant testing. First, EBV may trigger autoantibody production via molecular mimicry, analogous to MS [48,49]. Recent studies have identified neutralizing autoantibodies against IL-10 in IBD [50], potentially triggered by viral IL-10 expression through mimicry [51]. Additionally, EBV may enhance mutagenesis in B cells, amplifying the production of antibodies reactive to microbiota or self-antigens, which may contribute to disease development. Another possibility is that EBV transcription factors, such as EBNA-2, interact with disease-associated genetic loci, modifying gene transcription. Beyond EBV, individuals who developed CD also exhibited pre-diagnostic overrepresentation of antibody responses against other herpesviruses, including CMV and HSV-1/2, suggesting a broader role for herpesviruses in immune activation during IBD development. These findings reinforce the notion that longitudinal, unbiased serological profiling not only validates previously suspected associations (e.g., EBV in MS), but also uncovers novel, disease-specific patterns, thereby providing a framework to investigate disease-specific immune mechanisms and to facilitate comparisons across different disease contexts.

In addition to EBV, pre-diagnostic CD was also characterized by a strong overrepresentation of anti-flagellin antibody responses up to 10 years before diagnosis. While anti-flagellin reactivity is a well-established pre-diagnostic marker of CD (e.g., anti-CBir1, anti-Fla-X, and anti-A4-Fla2 antibodies) [3,5–7,9], our high-resolution analysis reveals the precise antigenic targets of these responses. In patients with established CD, we previously identified three distinct regions preferentially targeted within both the N- and C-terminal domains of flagellins [22]. However, in the current pre-diagnostic analysis, a single dominant cluster of antibody responses was detected, directed against the ‘hinge epitope’ region (nD0–$1_{p25–59}$) in the N-terminal domain. This region contains a prominent B cell epitope recently reported as a preferential target in both CD and healthy infants [44]. These highly overrepresented anti-flagellin antibodies were predominantly directed against *Lachnospiraceae* and were strongly associated with complicated CD at diagnosis. This observation suggests a potential shift from a homeostatic immune response to flagellins during infancy into a pathologic response later in life, aligning with recent observations [44]. Notably, these findings are highly concordant with recent observations from the GEM cohort, where elevated IgG responses against the same conserved *Lachnospiraceae* flagellin hinge epitope were shown to predict future CD in first-degree relatives [52]. These findings may conceptually align with elements of the hygiene hypothesis, which postulates that altered early-life microbial exposures can shape immune development and influence susceptibility to immune-mediated diseases, including IBD. Reduced microbial diversity or

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altered immune conditioning early in life could theoretically contribute to aberrant immune responses later in adulthood. However, our data do not directly address early-life exposures and should therefore not be interpreted as mechanistic support for this hypothesis, but rather as providing a framework for future investigation.

In individuals who developed CD and UC, antibody responses against antigens from potentially pathogenic, often encapsulated, bacteria, including *Streptococcus* (e.g., surface protein A), *Neisseria* (e.g., IgA1 protease), and *Haemophilus* (e.g., heme/hemopexin-binding protein HxuA) declined progressively towards diagnosis. This trend aligns with observations in patients with longstanding IBD [22] and suggests an impaired ability to mount immune responses against encapsulated bacteria, potentially reflecting broader immune dysregulation associated with increasing inflammation and/or disrupted immune tolerance in individuals developing IBD.

In contrast to prior work [3,5,10], the diagnosis of UC is preceded by a combination of antimicrobial, antiviral, and autoantibody signatures that included antibodies against a range of antigens, such as antimicrobial antibodies directed to *Streptococcus* (e.g., surface protein A), Bacteroidales (e.g., peptidase C10 from *Paraprevotella*), *Staphylococcus* (e.g., lipase 2), as well as antiviral and autoantibodies. Notably, we observed escalating trajectories of antibodies against MADD protein closer to UC diagnosis. MADD plays a key role in regulating inflammatory responses through the activation of mitogen-activated protein kinases (MAPKs) [53]. Dysregulation of MADD could lead to impaired lymphocyte cytotoxicity, as has been reported in case of complete MADD deficiency [54]. Pre-diagnostic autoantibodies against MADD may serve as a novel biomarker for pre-diagnostic UC.

Another finding delineated herein is the observation that differential antibody signatures exist for complicated CD already ~10 years prior to diagnosis. Similarly, our data demonstrate differential antibody signatures for disease location (e.g., ileal vs. colonic CD) prior to disease diagnosis, highlighting the potential to identify high-risk patients for the application of preventive strategies. These observations underscore that pre-diagnostic antibody repertoires not only reflect imminent disease, but can also stratify individuals by risk of disease phenotype, a concept that may be relevant to future precision prevention strategies in IBD. Interestingly, differential antibody signatures were comparable across several pre-diagnostic timepoints and in some instances even slightly more prominent pre-diagnosis than at the diagnostic timepoint. This may reflect increasing biological heterogeneity around diagnosis, potentially driven by inflammation, treatment initiation, or other peri-diagnostic factors that may obscure earlier immunological alterations. However, the analyses performed in this study should be interpreted as proof-of-concept that pre-diagnostic antibody repertoires contain discriminative signal rather than constituting finalized input for clinical prediction models. Future studies may focus on testing the predictive ability of these signatures formed into reduced antibody panels that retain high predictive performance while improving clinical feasibility, ideally using dedicated biomarker development studies and independent validation cohorts.

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The main strengths of this study pertain to the unique features of our study cohort, including its longitudinal design with multiple pre-diagnostic samples, the inclusion of an age-, sex-, race-, and timepoint-matched control group, a large sample size, and comprehensive disease information at diagnosis. Compared with previous analyses of antibody repertoires in population-based cohorts such as the Lifelines study [23], the present design further enables evaluation of temporal antibody trajectories across multiple pre-diagnostic timepoints and allows assessment of differential antibody signatures up to a decade before diagnosis. Equally important, this study demonstrates the feasibility of combining large pre-diagnostic biobanks with high-throughput serological profiling, a strategy that may also be applicable to the study of other immune-mediated conditions such as RA, lupus, or T1D. To our knowledge, this is the first study to apply massively parallel serology at scale in a pre-diagnostic cohort, thereby establishing a potential framework for similar analyses in other biobank-linked diseases.

Certain limitations of this study also warrant recognition. These include the relative uniqueness of our cohort consisting of active US military service. However, it should be noted that, due to its unique design, a comparable pre-diagnostic cohort like PREDICTS does not (yet) exist, which for now precluded external validation in an independent cohort. Our findings may have limited external generalizability with respect to the broader UC population, as the PREDICTS cohort predominantly consists of male participants (>85%) and elderly-onset IBD is underrepresented in this population. Although age and sex are known determinants of antibody epitope repertoires in population-level analyses, this demographic structure of PREDICTS precluded robust evaluation of age- or sex-specific effects. Furthermore, potential effects of initial treatment initiation around the time of diagnosis cannot be fully excluded, although considered highly unlikely as initiation of such treatment typically affects deployment and active duty status. Another limitation is that detailed information on geographic location or seasonal timing of sample collection was not available within the DMSS-DoDSR. Although cases and controls were matched on the calendar year of the diagnostic timepoint to partially account for temporal effects, regional or seasonal differences in infectious exposures (which may influence antibody repertoires) could not be evaluated. Additionally, while our findings strongly suggest potential microbial and non-microbial triggers contributing to IBD pathogenesis, causal inference cannot be made without supportive mechanistic data. Finally, it is estimated that PhIP-Seq may only detect ~10% of the entire antigenic space [16,29], although there is no current technology that can effectively surpass the high-throughput potential of PhIP-Seq. While PhIP-Seq does not yet cover the full antigenic space, its breadth, scalability, and compatibility with archived serum samples make it uniquely suited for discovery efforts in large prospective biobanks.

In summary, this study presents a novel resource of immunological information spanning a period of up to 10 years before IBD diagnosis, which highlights the vastness, diversity, and specificity of early immunological changes preceding IBD development. More broadly, our work illustrates how large-scale serology can be leveraged to decipher prodromal immune changes in IBD, thereby opening avenues for early detection and improved risk stratification in IBD. Our findings suggest infectious triggers as potential drivers of the disease and/or early disruptions in host immunity, offering novel insights into IBD pathogenesis, and may ultimately inform future strategies aimed at mitigating or preventing IBD, as has

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been recently demonstrated in type 1 diabetes. Taken together, these data demonstrate how combining pre-diagnostic biobanks with high-throughput serology yields unprecedented insights into the prodromal immune system, offering a promising strategy to study early disease biology in IBD and to support future efforts aimed at prevention.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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Data and code availability

All requests for raw and analyzed data and materials will be promptly reviewed by the corresponding authors and study team. Raw sequencing files have been deposited in the Harvard Dataverse repository and will be made publicly available upon publication. Currently, they are accessible through the following private link: https://dataverse.harvard.edu/previewurl.xhtml?token=4af68e23-6ae8-49d3-9e0e-81307a114479. The code developed that is central to the main work described in this manuscript has been published. This antigen/peptide creation code can be accessed through the respective article by Vogl et al., *Nature Medicine*, 2021: "Population-wide diversity and stability of serum antibody epitope repertoires against human microbiota." *Nat Med* 27, 1442–1450 (2021), doi: 10.1038/s41591-021-01409-3 or directly via https://github.com/erans99/PhIPSeq_external. Custom code used for basic statistical comparisons will be made publicly available upon publication in Zenodo.

References

1. Torres J, Mehandru S, Colombel JF, Peyrin-Biroulet L. Crohn's disease. Lancet. 2017;389:1741–1755. doi:10.1016/S0140-6736(16)31711-1.
2. Ungaro R, Mehandru S, Allen PB, Peyrin-Biroulet L, Colombel JF. Ulcerative colitis. Lancet. 2017;389:1756–1770. doi:10.1016/S0140-6736(16)32126-2.
3. Torres J, Petralia F, Sato T, et al. Serum biomarkers identify patients who will develop inflammatory bowel diseases up to 5 years before diagnosis. Gastroenterology. 2020;159:96–104. doi:10.1053/j.gastro.2020.03.007.
4. de Souza HSP, Fiocchi C, Iliopoulos D. The IBD interactome: an integrated view of aetiology, pathogenesis and therapy. Nat Rev Gastroenterol Hepatol. 2017;14:739–749. doi:10.1038/nrgastro.2017.110.

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5. Israeli E, Grotto I, Gilburd B, et al. Anti-Saccharomyces cerevisiae and antineutrophil cytoplasmic antibodies as predictors of inflammatory bowel disease. Gut. 2005;54:1232–1236. doi:10.1136/gut.2004.060228.
6. van Schaik FD, Oldenburg B, Hart AR, et al. Serological markers predict inflammatory bowel disease years before the diagnosis. Gut. 2013;62:683–688. doi:10.1136/gutjnl-2012-302717.
7. Choung RS, Princen F, Stockfisch TP, et al. Serologic microbial associated markers can predict Crohn's disease behaviour years before disease diagnosis. Aliment Pharmacol Ther. 2016;43:1300–1310. doi:10.1111/apt.13641.
8. Alexander KL, Zhao Q, Reif M, et al. Human microbiota flagellins drive adaptive immune responses in Crohn's disease. Gastroenterology. 2021;161:522–535.e6. doi:10.1053/j.gastro.2021.03.064.
9. Lee SH, Turpin W, Espin-Garcia O, et al. Anti-microbial antibody response is associated with future onset of Crohn's disease independent of biomarkers of altered gut barrier function, subclinical inflammation, and genetic risk. Gastroenterology. 2021;161:1540–1551. doi:10.1053/j.gastro.2021.07.009.
10. Livanos AE, Dunn A, Fischer J, et al. Anti-integrin αvβ6 autoantibodies are a novel biomarker that antedate ulcerative colitis. Gastroenterology. 2023;164:619–629. doi:10.1053/j.gastro.2022.12.042.
11. Pasolli E, Asnicar F, Manara S, et al. Extensive unexplored human microbiome diversity revealed by over 150,000 genomes from metagenomes spanning age, geography, and lifestyle. Cell. 2019;176:649–662. doi:10.1016/j.cell.2019.01.001.
12. Sterlin D, Fadlallah J, Slack E, Gorochov G. The antibody/microbiota interface in health and disease. Mucosal Immunol. 2020;13:3–11. doi:10.1038/s41385-019-0192-y.
13. Bourgonje AR, Vogl T, Segal E, Weersma RK. Antibody signatures in inflammatory bowel disease: current developments and future applications. Trends Mol Med. 2022;28:693–705. doi:10.1016/j.molmed.2022.05.004.
14. Larman HB, Zhao Z, Laserson U, et al. Autoantigen discovery with a synthetic human peptidome. Nat Biotechnol. 2011;29:535–541. doi:10.1038/nbt.1856.
15. Mohan D, Wansley DL, Sie BM, et al. PhIP-Seq characterization of serum antibodies using oligonucleotide-encoded peptidomes. Nat Protoc. 2018;13:1958–1978. doi:10.1038/s41596-018-0025-6.
16. Vogl T, Klompus S, Leviatan S, et al. Population-wide diversity and stability of serum antibody epitope repertoires against human microbiota. Nat Med. 2021;27:1442–1450. doi:10.1038/s41591-021-01409-3.
17. Mina MJ, Kula T, Leng Y, et al. Measles virus infection diminishes preexisting antibodies that offer protection from other pathogens. Science. 2019;366:599–606.
18. Xu GJ, Kula T, Leng Y, et al. Measles virus infection diminishes preexisting antibodies that offer protection from other pathogens. Science. 2015;348:aaa0698. doi:10.1126/science.aaa0698.
19. Zamecnik CR, Sowa GM, Abdelhak A, et al. An autoantibody signature predictive for multiple sclerosis. Nat Med. 2024;30:1300–1308. doi:10.1038/s41591-024-02938-3.
20. Bjornevik K, Cortese M, Healy BC, et al. Longitudinal analysis reveals high prevalence of Epstein-Barr virus associated with multiple sclerosis. Science. 2022;375:296–301. doi:10.1126/science.abj8222.
21. Larman HB, Laserson U, Querol L, et al. PhIP-Seq characterization of autoantibodies from patients with multiple sclerosis, type 1 diabetes and rheumatoid arthritis. J Autoimmun. 2013;43:1–9. doi:10.1016/j.jaut.2013.01.013.
22. Bourgonje AR, Andreu-Sánchez S, Vogl T, et al. Phage-display immunoprecipitation sequencing of the antibody epitope repertoire in inflammatory bowel disease reveals distinct antibody signatures. Immunity. 2023;56:1393–1409.e6. doi:10.1016/j.immuni.2023.04.017.
23. Bourgonje AR, Andreu-Sánchez S, Gacesa R, et al. Crohn's disease and ulcerative colitis exhibit pre-diagnostic antibody signatures with shared and divergent changes towards disease onset. Gut. 2025. doi:10.1136/gutjnl-2024-334362.
24. Nandy A, Petralia F, Porter C, et al. Epstein-Barr virus (EBV) exposure precedes Crohn's disease development. Gastroenterology. 2025;S0016–5085(25)00411–1. doi:10.1053/j.gastro.2025.01.247.

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25. Porter CK, Riddle MS, Gutierrez RL, et al. Cohort profile of the PRoteomic Evaluation and Discovery in an IBD Cohort of Tri-service Subjects (PREDICTS) study: rationale, organization, design, and baseline characteristics. Contemp Clin Trials Commun. 2019;14:100345. doi:10.1016/j.conctc.2019.100345.
26. Hutfless S, Jasper RA, Tilak A, Ghosh T, Kedia S, Liu S, et al. A Systematic Review of Crohn's Disease Case Definitions in Administrative or Claims Databases. Inflamm Bowel Dis. 2023;29(5):705–715. doi: 10.1093/ibd/izac131.
27. Thirumurthi S, Chowdhury R, Richardson P, Abraham NS. Validation of ICD-9-CM diagnostic codes for inflammatory bowel disease among veterans. Dig Dis Sci. 2010;55(9):2592–8. doi: 10.1007/s10620-009-1074-z.
28. Hou JK, Tan M, Stidham RW, Colozzi J, Adams D, El-Serag H, Waljee AK. Accuracy of diagnostic codes for identifying patients with ulcerative colitis and Crohn's disease in the Veterans Affairs Health Care System. Dig Dis Sci. 2014;59(10):2406–10. doi:10.1007/s10620-014-3174-7.
29. Leviatan S, Vogl T, Klompus S, Kalka IN, Weinberger A, Segal E. Allergenic food protein consumption is associated with systemic IgG antibody responses in non-allergic individuals. Immunity. 2023;55:2454–2469.e6. doi:10.1016/j.immuni.2022.11.004.
30. Klompus S, Leviatan S, Vogl T, et al. Cross-reactive antibodies against human coronaviruses and the animal coronavirome suggest diagnostics for future zoonotic spillovers. Sci Immunol. 2021;6:eabe9950. doi:10.1126/sciimmunol.abe9950.
31. Zeevi D, Korem T, Zmora N, et al. Personalized nutrition by prediction of glycemic responses. Cell. 2015;163:1079–1094. doi:10.1016/j.cell.2015.11.001.
32. Lebeer S, Bron PA, Marco ML, et al. Identification of probiotic effector molecules: present state and future perspectives. Curr Opin Biotechnol. 2018;49:217–223. doi:10.1016/j.copbio.2017.10.007.
33. Palm NW, de Zoete MR, Cullen TW, et al. Immunoglobulin A coating identifies colitogenic bacteria in inflammatory bowel disease. Cell. 2014;158:1000–1010. doi:10.1016/j.cell.2014.08.006.
34. Chen L, Zheng D, Liu B, Yang J, Jin Q. VFDB 2016: hierarchical and refined dataset for big data analysis—10 years on. Nucleic Acids Res. 2016;44:D694–D697. doi:10.1093/nar/gkv1239.
35. Bourgonje AR, Hörstke NV, Fehringer M, Innocenti G, Vogl T. Systemic antibody responses against gut microbiota flagellins implicate shared and divergent immune reactivity in Crohn's disease and chronic fatigue syndrome. Microbiome. 2024;12:141. doi:10.1186/s40168-024-01858-1.
36. Radauer C, Nandy A, Ferreira F, et al. Update of the WHO/IUIS Allergen Nomenclature Database based on analysis of allergen sequences. Allergy. 2014;69:413–419. doi:10.1111/all.12348.
37. Goodman RE, Ebisawa M, Ferreira F, et al. AllergenOnline: a peer-reviewed, curated allergen database to assess novel food proteins for potential cross-reactivity. Mol Nutr Food Res. 2016;60:1183–1198. doi:10.1002/mnfr.201500769.
38. Mari A, Scala E, Palazzo P, Ridolfi S, Zennaro D, Carabella G. Bioinformatics applied to allergy: allergen databases, from collecting sequence information to data integration. The Allergome platform as a model. Cell Immunol. 2006;244:97–100. doi:10.1016/j.cellimm.2007.02.012.
39. Radauer C, Bublin M, Wagner S, Mari A, Breiteneder H. Allergens are distributed into few protein families and possess a restricted number of biochemical functions. J Allergy Clin Immunol. 2008;121:847–857.e7. doi:10.1016/j.jaci.2008.01.025.
40. Ivanciuc O, Schein CH, Braun W. SDAP: database and computational tools for allergenic proteins. Nucleic Acids Res. 2003;31:359–362. doi:10.1093/nar/gkg010.
41. Fitzsimmons CM, Falcone FH, Dunne DW. Helminth allergens, parasite-specific IgE, and its protective role in human immunity. Front Immunol. 2014;5:61. doi:10.3389/fimmu.2014.00061.
42. Vita R, Overton JA, Greenbaum JA, et al. The immune epitope database (IEDB) 3.0. Nucleic Acids Res. 2015;43:D405–D412. doi:10.1093/nar/gku938.
43. Kumar S, Stecher G, Li M, Knyaz C, Tamura K. MEGA X: molecular evolutionary genetics analysis across computing platforms. Mol Biol Evol. 2018;35:1547–1549. doi:10.1093/molbev/msy096.

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44. Zhao Q, Duck LW, Killian JT Jr, et al. Crohn's patients and healthy infants share immunodominant B cell response to commensal flagellin peptide epitopes. Gastroenterology. 2024;167:1415–1428. doi:10.1053/j.gastro.2024.08.015.
45. Vietzen H, Kühner LM, Berger SM, et al. Accumulation of Epstein-Barr virus-induced cross-reactive immune responses is associated with multiple sclerosis. J Clin Invest. 2024;134:e184481. doi:10.1172/JCI184481.
46. Jog NR, McClain MT, Heinlen LD, et al. Epstein-Barr virus nuclear antigen 1 (EBNA-1) peptides recognized by adult multiple sclerosis patient sera induce neurologic symptoms in a murine model. J Autoimmun. 2020;106:102332. doi:10.1016/j.jaut.2019.102332.
47. Ebert AC, Harper S, Vestergaard MV, Mitchell W, Jess T, Elmahdi R. Risk of inflammatory bowel disease following hospitalisation with infectious mononucleosis: nationwide cohort study from Denmark. Nat Commun. 2024;15:8383. doi:10.1038/s41467-024-52195-8.
48. Lanz TV, Brewer RC, Ho PP, et al. Clonally expanded B cells in multiple sclerosis bind EBV EBNA1 and GlialCAM. Nature. 2022;603:321–327. doi:10.1038/s41586-022-04432-7.
49. Thomas OG, Bronge M, Tengvall K, et al. Cross-reactive EBNA1 immunity targets alpha-crystallin B and is associated with multiple sclerosis. Sci Adv. 2023;9:eadg3032. doi:10.1126/sciadv.adg3032.
50. Griffin H, Ceron-Gutierrez L, Gharahdaghi N, et al. Neutralizing autoantibodies against interleukin-10 in inflammatory bowel disease. N Engl J Med. 2024;391:434–441. doi:10.1056/NEJMoa2312302.
51. Jog NR, Chakravarty EF, Guthridge JM, James JA. Epstein-Barr virus interleukin 10 suppresses anti-inflammatory phenotype in human monocytes. Front Immunol. 2018;9:2198. doi:10.3389/fimmu.2018.02198.
52. Wu RY, Xue M, Zhao Q, et al. Serum IgG Response to a Conserved Domain of Commensal Flagellins Predicts Future Risk of Crohn's Disease in First-degree Relatives. Clin Gastroenterol Hepatol. 2026:S1542–3565(25)01031–6. doi: 10.1016/j.cgh.2025.12.006.
53. Branger J, van den Blink B, Weijer S, et al. Anti-inflammatory effects of a p38 mitogen-activated protein kinase inhibitor during human endotoxemia. J Immunol. 2002;168:4070–4077. doi:10.4049/jimmunol.168.8.4070.
54. Schütze K, Groß M, Cornils K, et al. MAP kinase activating death domain deficiency is a novel cause of impaired lymphocyte cytotoxicity. Blood Adv. 2023;7:1531–1535. doi:10.1182/bloodadvances.2022008195.

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Key messages

What is already known on this topic

- The preclinical phase of inflammatory bowel disease (IBD) remains poorly characterized, limiting opportunities for early disease detection.

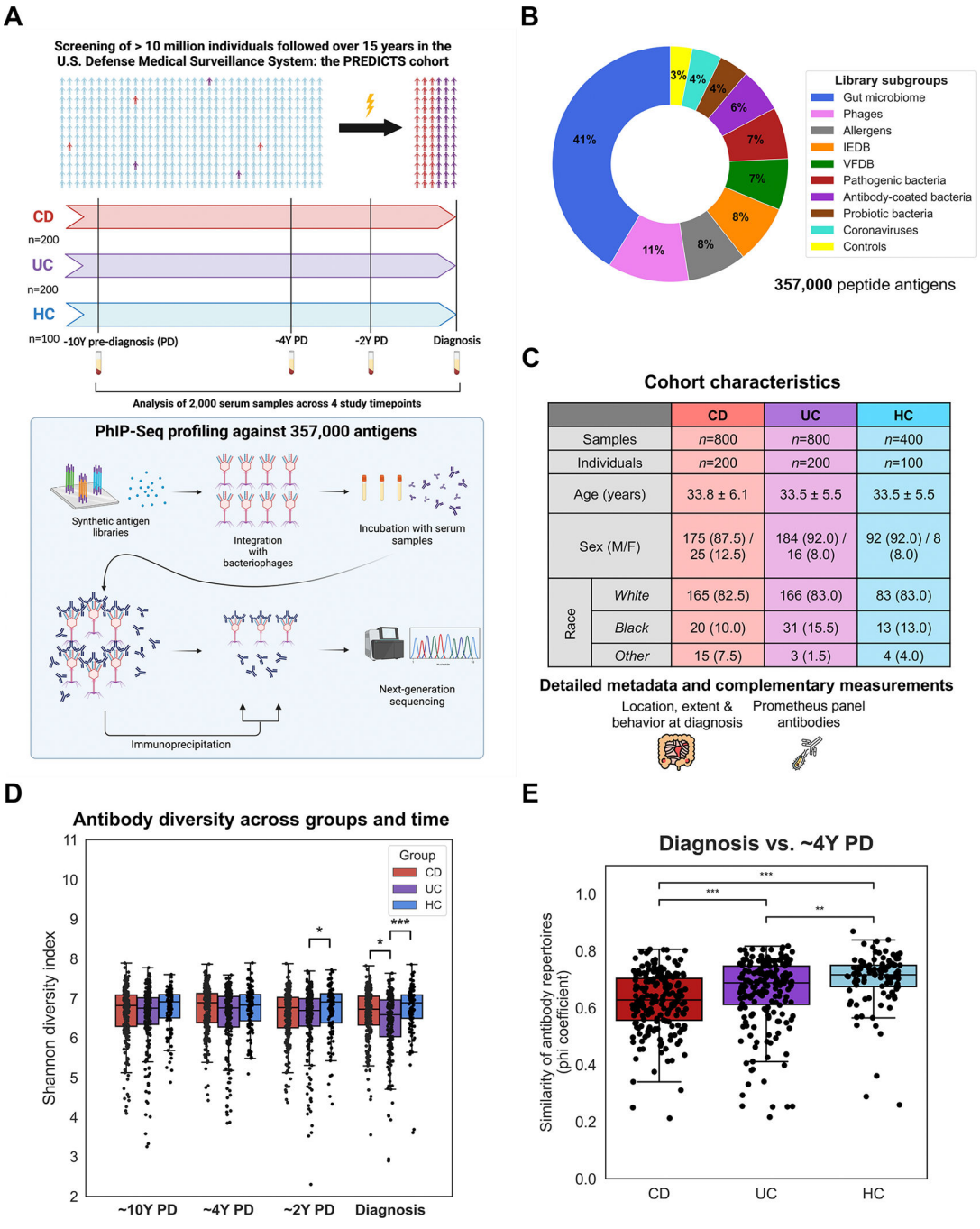
What this study adds

- Antibody repertoires against 357,000 antigens were profiled longitudinally in the pre-clinical phase of IBD, revealing distinct antibody signatures up to 10 years before diagnosis, especially against Epstein-Barr virus and bacterial flagellins.

How this study might affect research, practice or policy

- These results provide novel insights into early immunological perturbations preceding IBD, highlighting dynamic antibacterial and antiviral antibody responses and supporting hypotheses of molecular mimicry and immune dysregulation as early events in disease pathogenesis.
- By systematically mapping longitudinal antibody trajectories in the pre-diagnostic setting, this study offers a framework to investigate early pathogenic mechanisms and to refine future strategies for risk stratification in IBD.

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		CD	UC	HC
Samples		*n*=800	*n*=800	*n*=400
Individuals		*n*=200	*n*=200	*n*=100
Age (years)		33.8 ± 6.1	33.5 ± 5.5	33.5 ± 5.5
Sex (M/F)		175 (87.5) / 25 (12.5)	184 (92.0) / 16 (8.0)	92 (92.0) / 8 (8.0)
Race	*White*	165 (82.5)	166 (83.0)	83 (83.0)
	Black	20 (10.0)	31 (15.5)	13 (13.0)
	Other	15 (7.5)	3 (1.5)	4 (4.0)

Figure 1. PhIP-Seq-based characterization of antibody repertoires.

(**A**) PhIP-Seq was performed to profile antibody epitope repertoires in longitudinal serum samples available from participants in the PREDICTS cohort developing CD (*n*=200, red), UC (*n*=200, purple) or not developing IBD (HC, blue, *n*=100). Serum samples were taken at ~10Y, ~4Y, and ~2Y prior to IBD diagnosis (PD). (**B**) Pie chart visualization of the composition of the antigen libraries [16,22,29,30]. (**C**) Cohort characteristics at time of diagnosis. (**D**) Boxplots showing Shannon diversity index distributions across samples from individuals who developed CD (red), UC (purple) or no IBD (HC, blue) at three

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PD timepoints and at diagnosis. (**E**) Boxplot showing overall longitudinal stability of antibody epitope repertoires between ~4Y PD and diagnosis for individuals developing CD, UC and matched HCs. Shown on the y-axis are the distributions of φ-coefficients. In panels **D** and **E** lower and upper boundaries of the boxes represent the 25th and 75th percentiles, with whiskers extending 1.5 times the interquartile range (IQR). Distributions were compared using pairwise Wilcoxon-Mann-Whitney *U*-tests with *P*-values corrected for multiple comparisons. **P*<0.05; ***P*<0.01; ****P*<0.001.

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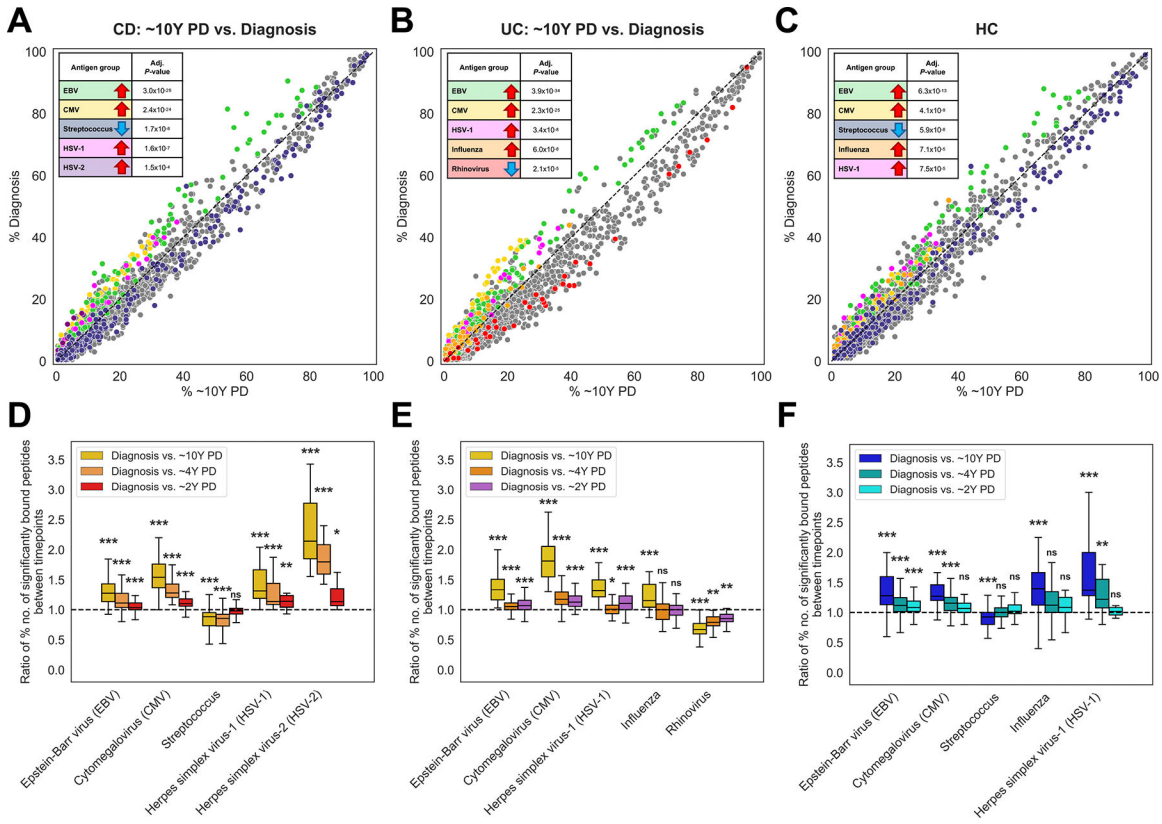


Figure 2. Longitudinal analysis of antibody epitope repertoires at ~10Y PD vs. diagnosis reveals antigen-specific changes in antibody responses with increased antibody reactivity against herpesviruses and decreased reactivity against encapsulated and potentially pathogenic bacteria and respiratory viruses towards IBD diagnosis ("within group" comparisons).

(**A-C**) Scatterplots showing the frequency (%) of antibody-bound peptides (every dot representing one antibody) at the farthest pre-diagnostic timepoint (~10Y PD, x-axis) and at time of diagnosis (y-axis) of CD (**A**), UC (**B**), or within matched HC samples (**C**). Dots are colored based on the top five significantly changing antigen subgroups (see **Tables S3-S5** for full lists of significantly changing groups) between ~10Y PD and diagnosis. Table insets show the corresponding directionality (upward-facing red arrows: higher frequency at diagnosis; downward-facing blue arrows: lower frequency at diagnosis) and adjusted *P*-values (ranked in descending order of significance) of the top five significantly changing antigen subgroups between ~10Y PD and diagnosis (**Tables S3-S5**) in individuals who developed CD, UC, and in matched HCs. *P*-values are derived from Wilcoxon-Mann-Whitney *U*-tests comparing frequency ratios for the antigen subgroups against the rest of the library, adjusted for multiple comparisons. (**D-F**) Boxplots demonstrating the ratios of frequencies (%) of antibody-bound peptides, classified within the top five changing antigen subgroups in individuals developing CD (**D**), UC (**E**), and matched HC samples (**F**), separated based on study timepoint (from left-to-right, samples obtained at the ~10Y, ~4Y, and ~2Y PD timepoints, compared to samples obtained at the diagnostic timepoint). Lower and upper boundaries of the boxes represent the 25th and 75th percentiles, with whiskers extending 1.5 times the interquartile range (IQR). Asterisks denote significantly different frequency ratios (*P<0.05; **P<0.01; ***P<0.001, Wilcoxon-Mann-Whitney *U*-

tests comparing frequency ratios for the antigen subgroups against the rest of the library, with *P*-values adjusted for multiple comparisons).

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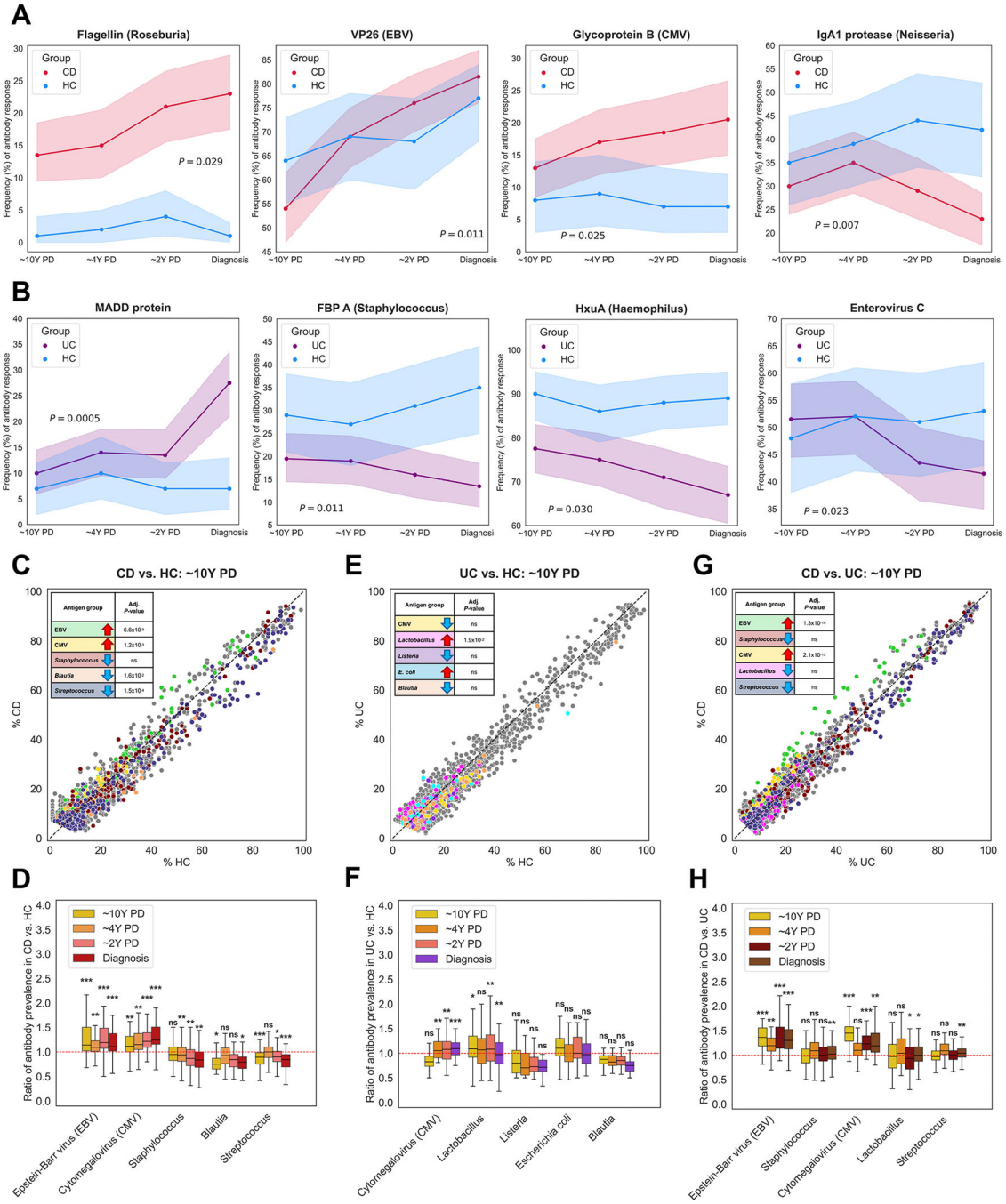


Figure 3. Distinct antimicrobial, antiviral, and autoantibody responses are detected up to 10 years before IBD diagnosis.

(**A, B**) Representative examples of longitudinal trajectories of antibody responses significantly changing over time (from ~10Y PD up until diagnosis) in individuals who developed CD (**A**) and UC (**B**) relative to matched HCs. Plots demonstrate frequencies (%) of antibody responses at each study timepoint (y-axis) that exhibited significant (filtered on nominal significance [$P<0.05$] and a cumulative percentual difference of >20%) interactions between groups (CD or UC vs. HC) and time (~10Y, ~4Y, or ~2Y PD vs. diagnosis). *P*-

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values shown represent *P*-values for *interaction* as derived from generalized logistic mixed-effects models (**Tables S21-S22**). Confidence bands represent 95% confidence intervals following bootstrap resampling with 1,000 iterations. (**C,E,G**) Scatterplots showing the frequency (%) of antibody-bound peptides (every dot represents one antibody) in HCs (x-axis) and in individuals who developed CD (**C**) or UC (**E**) (y-axis) or UC and CD (**G**) (x- and y-axis, respectively) at the farthest pre-diagnostic timepoint (~10Y PD). Dots are colored based on the top five differentially abundant antigen subgroups per comparison based on *averaged* adjusted *P*-values across all four study timepoints (see **Table S23** for full overview). Table inserts show the corresponding directionality (upward-facing red arrows: higher frequency in CD (panels **C,G**) or UC (panel **E**); downward-facing blue arrows: lower frequency in CD (**C,G**) or UC (**D**)) and adjusted *P*-values (here only shown for the ~10Y PD timepoint) of the top five differentially abundant antigen subgroups between CD and HC (**C**), UC and HC (**E**), and CD and UC (**G**) (**Table S23**). *P*-values are derived from Wilcoxon-Mann-Whitney *U*-tests comparing frequency ratios of antibodies annotated for the antigen subgroups against those of the rest of the library, adjusted for multiple comparisons. (**D,F,H**) Boxplots showing the ratios of antibody-bound peptide frequencies (%) among the top five differentially abundant antigen subgroups (based on averaged *P*-values across timepoints) for each comparison, i.e. individuals who developed CD vs. HC (**D**), individuals who developed UC vs. HC (**F**), and individuals who developed CD vs. UC (**H**). Study timepoints are labeled in chronological order. Antigen subgroups are ranked from left-to-right in descending order of statistical significance (averaged across all timepoints). Lower and upper boundaries of the boxes represent the 25th and 75th percentiles, with whiskers extending 1.5 times the interquartile range (IQR). Asterisks denote significantly different frequency ratios (**P*<0.05; ***P*<0.01; ****P*<0.001, Wilcoxon-Mann-Whitney *U*-tests comparing frequency ratios for the antigen subgroups against the rest of the library, with *P*-values adjusted for multiple comparisons).

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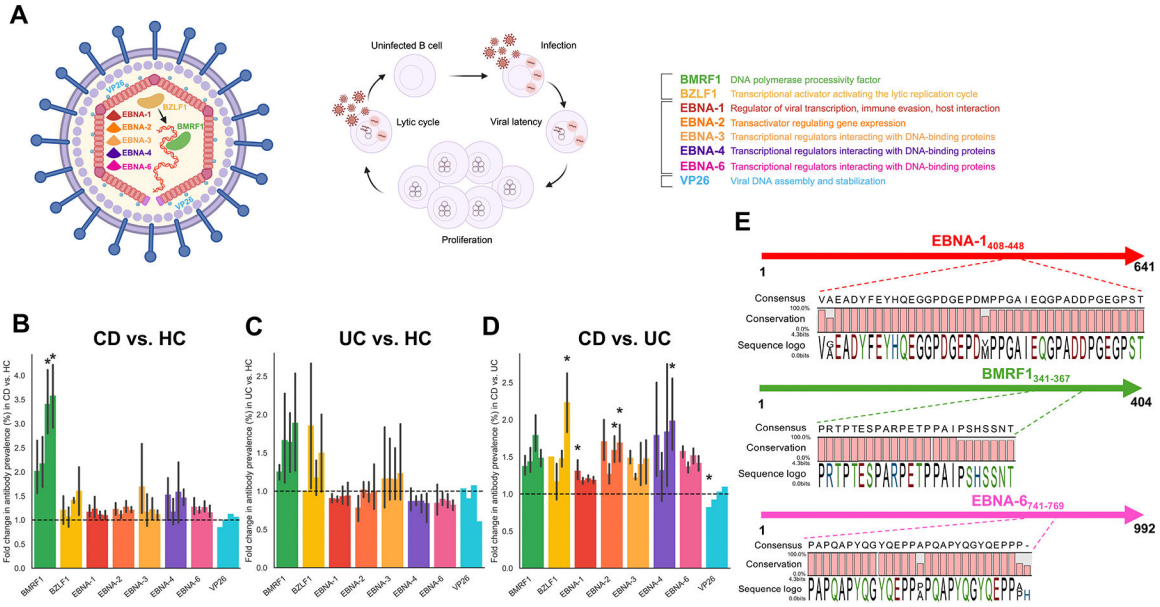


Figure 4. Antibody responses against Epstein-Barr virus (EBV)-antigens are overrepresented up to 10 years before the diagnosis of CD and directed to distinct antigen sequence motifs.

(**A**) Schematic illustration displaying the basic structural elements of the EBV viral particle, including a simplified representation of the key antigens bound by antibodies in study participants. (**B-D**) Bar plots showing fold changes in antibody-bound peptide frequencies (%) of all EBV-directed antibodies that were found to be differentially abundant (see **Table S38**) in all comparisons across all four study timepoints, including individuals developing CD vs. HC (**B**), UC vs. HC (**C**), and CD vs. UC (**D**). Each individual bar represents the fold change per study timepoint in chronological order, i.e. the four bars from left-to-right represent ~10Y PD, ~4Y PD, ~2Y PD, and diagnosis. Asterisks (*) indicate the presence of significantly differentially abundant anti-EBV antibodies (**Table S38**) per timepoint annotated to the antigens shown on the x-axis. (**E-G**) Full sequence alignment of all differentially abundant anti-EBV antibodies yielded three distinct clusters sharing conserved sequence motifs putatively bound by cross-reactive antibody responses against EBV, specifically for EBNA-1 (**E**), BMRF1 (**F**), and EBNA-6 (**G**) antigens.

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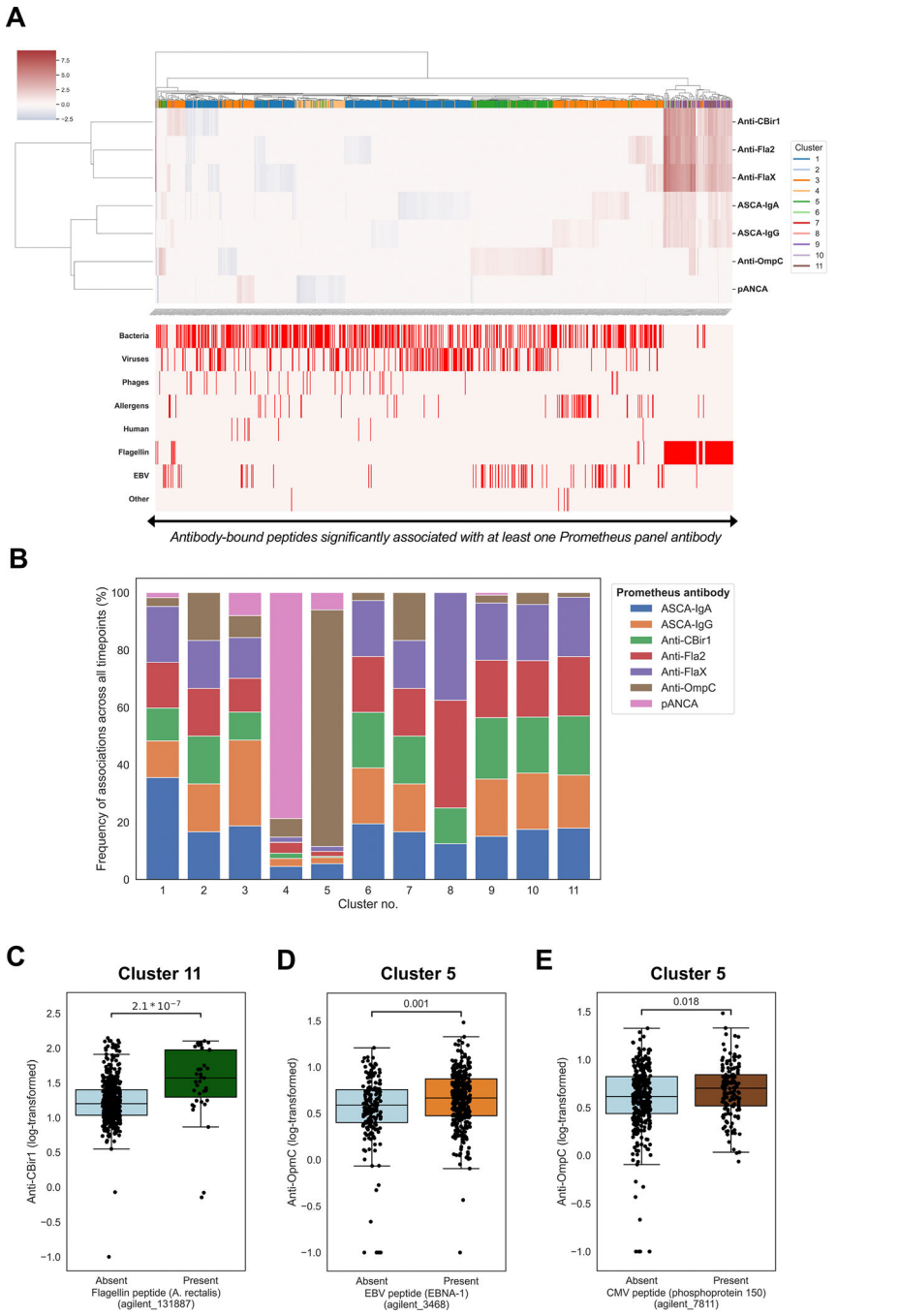


Figure 5. Established serological biomarkers for diagnosis (Prometheus panel antibodies) exhibit shared and divergent co-occurrences with PhIP-Seq-derived antibody reactivities.

(**A**) Heatmap visualizing associations between Prometheus panel antibodies (ASCA-IgG, ASCA-IgA, anti-flagellin antibodies anti-CBir1, anti-FlaX, and anti-Fla2, and anti-*E. coli* outer membrane protein C, anti-OmpC, shown on the y-axis) and PhIP-Seq-derived antibody-bound peptides (x-axis level) after hierarchical clustering (using the average linkage method), showing both shared and distinct clusters of antibody-bound peptides either positively (red color) or negatively (blue color) associated with each Prometheus

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panel antibody (upper heatmap). Only antibody-bound peptides associated with at least one Prometheus panel antibody for at least one study timepoint (with nominal $P<0.01$) were selected for clustering. Upper heatmap coloring is based on coefficient estimates derived from logistic regression analyses associating Prometheus panel antibodies with antibody-bound peptides at each individual timepoint. Column-level labeling represents the 11 different clusters of antibody-bound peptides identified after hierarchical clustering. Lower heatmap depicts the major taxonomical annotations of antibody-bound peptides including flagellin and EBV-annotated peptides which commonly occurred in specific clusters. (**B**) Stacked bar plot visualizing the frequencies (%) of associations across all study timepoints accounted for by each Prometheus panel antibody within each cluster of antibody-bound peptides (x-axis). (**C-E**) Representative examples of boxplots showing positive associations between PhIP-Seq-derived antibody-bound peptides and Prometheus panel antibodies, including (**C**) cluster 3 that is dominated by EBV-derived antibody-bound peptides co-occurring with anti-OmpC antibodies, (**D**) cluster 5 in which CMV-associated peptides (here: phosphoprotein 150) co-occur with anti-OmpC antibodies, and (**E**) cluster 11 characterized by co-occurrence of antibody-bound flagellin peptides (here: *Agathobacter rectalis* flagellin) and anti-CBir1 antibodies.

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Table 1.

Demographic and clinical characteristics of the study population at time of diagnosis.

Variable	CD	UC	HC
No. of samples	*n*=800	*n*=800	*n*=400
No. of individuals	*n*=200	*n*=200	*n*=100
Age at diagnosis (years) (mean ± SD)	33.0 ± 6.1	33.5 ± 5.5	33.5 ± 5.5
Male/female sex, *n* (%)	175 (87.5) / 25 (12.5)	184 (92.0) / 16 (8.0)	92 (92.0) / 8 (8.0)
Race, n (%)			
White	165 (82.5)	166 (83.0)	83 (83.0)
Black	20 (10.0)	31 (15.5)	13 (13.0)
Other	15 (7.5)	3 (1.5)	4 (4.0)
Disease location, *n* (%)			
L1 (ileal)	37 (18.5)	-	-
L2 (colonic)	44 (22.0)	-	-
L3 (ileocolonic)	65 (32.5)	-	-
Unknown	54 (27.0)	-	-
Disease extent, *n* (%)			
E1 (proctitis)	-	29 (14.5)	-
E2 (left-sided colitis)	-	43 (21.5)	-
E3 (extensive colitis)	-	74 (37.0)	-
Unknown	-	54 (27.0)	
Disease behavior, *n* (%)			
B1 (non-stricturing, non-penetrating)	146 (73.0)	-	-
B2 (stricturing)	29 (14.5)	-	-
B3 (penetrating)	19 (9.5)	-	-
Unknown	6 (3.0)		
Perianal disease, *n* (%)	16 (8.0)	-	-

Abbreviations: CD, Crohn's disease; HC, healthy controls; SD, standard deviation; UC, ulcerative colitis.